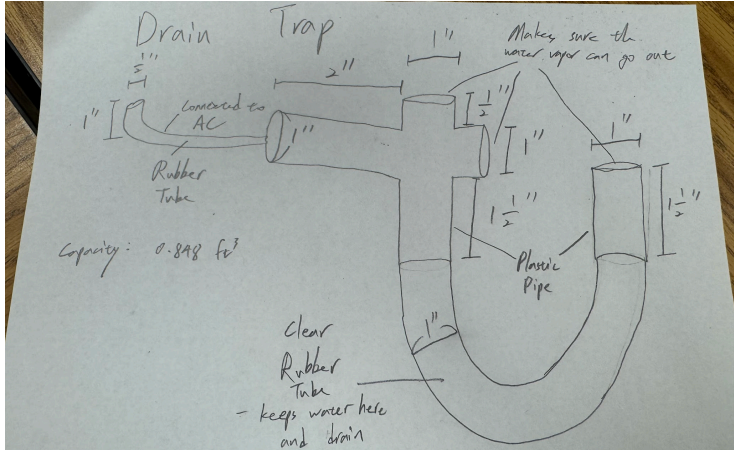


ELEMENT D

D1. PROCESS FOR GENERATING & COMPARING POSSIBLE SOLUTIONS

Requirements:	Drain trap	Collecting Tub	Drain Pan	Rubber Tubing
500 mL capacity	The device provides a 0.848 ft ³ area and, when calculated, equals 24000 mL.	Has an area where it collects water; area of 1 ft ³ , allowing it to hold 28316 mL.	Has an area where it collects water; area of 0.8 ft ³ , allowing it to hold 22000 mL.	This device does not have a storage unit to hold the water, leaving the water to drip through the other side and not having a 500 mL capacity.
Under 2 ft ² contact surface with water	(0.25 in x π x length of the pipe)	(only has an area of 1 ² , not covering the whole AC area).	Has an area of (more than 4 ft ²)	(at least 1 in x π x length of the pipe {at least 3 ft})
Able to replace in a min	The device does not need to be deconstructed, only attached to the AC. Allowing it to be connected and disconnected easily.	Due to the PVC in the legs, it will not reach the 1 minute mark, taking about 3 to 5 minutes to separate the PVC pipes.	The water may be heavy to carry but can be easily dumped and stored while still fully in contact.	Can be easily disconnected and connected to the AC without further steps needed to slow the process down.
Must hold 90% of condensation, minimize leakage	Possible leakage if there aren't caps designed for the pipes to close it before getting full.	Can hold most of the 90% of condensation, but it is possible for leakage if the filter is blocked with debris and relying on the overflow drainage, which will leak the water out.	An open container: leakage is slightly possible when the water drops onto the water puddle.	Can not hold condensed water, leading the water to the other end of the pipe.
No blockage throughout the design	The device has some areas where the debris can enter due to the pipes having a hole in it to let out water vapor.	Debris can easily enter through the top of the tub, allowing leaves or other debris to clog the main source of where the water will exit, but does have an emergency exit point if it can not hold the condensation.	The device doesn't have a cover but wouldn't cause too much trouble because there isn't a passage for the debris to block.	No debris will be able to clog it due to the tubing being connected to the AC and the designated area.

Concept 1: Drain Trap



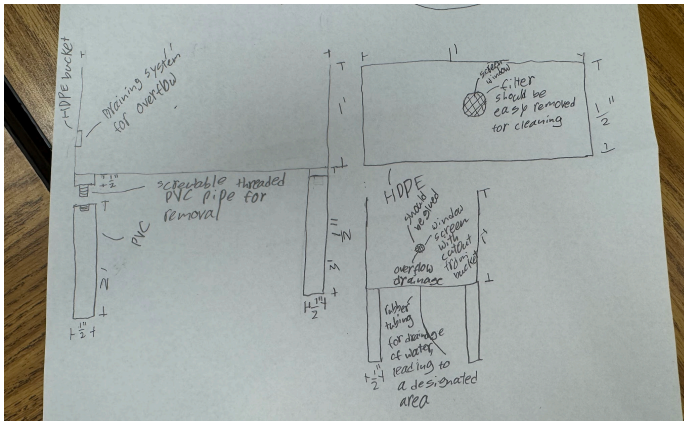
Description:

The drain trap is connected to the AC units with a 0.5 in diameter rubber tube. Then water will enter through a 2" long tilted plastic pipe, then enter the 1 in diameter U-shaped rubber tube, which would trap the condensate and drain it. All plastic pipes connected to the rubber tube will be open to ensure the release of water vapor so water doesn't overflow the device.

Advantages: The connection to the AC allows no water to be spilled, and with the addition of the pipes connected, it makes it less likely for water to leak and drip onto the walkway. And the opening outside of the pipes allows water vapor to exit, releasing pressure.

Disadvantages: The opening on the top allows debris to enter through the top; leaves and tiny debris have a larger chance of clogging the device and not letting water flow through the device. There is very limited capacity for the water to be collected; there is also no extra tube for water to be led to another storage or a designated area for the water to lead out.

Concept 2: Collecting Tub



Description:

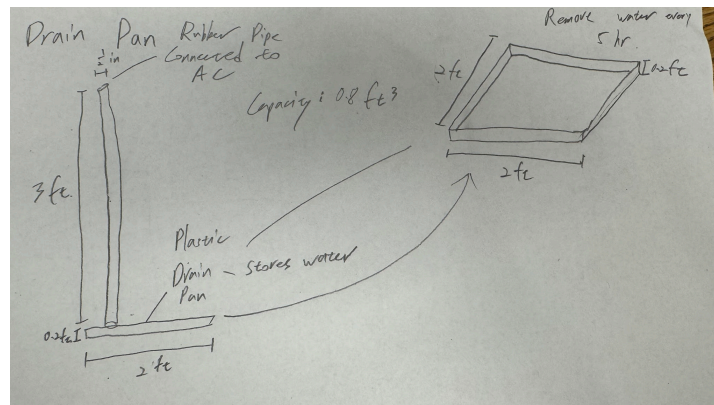
The Square Bucket design allows it to sit underneath the AC and collect water drops that the AC may leak, the bucket dimension will be 1' long by ½" wide by 1' tall and made from HDPE material. There are PVC pipes

underneath the bucket that are screwed to a threaded PVC pipe for easy removal. The threaded PVC pipe will be connected to the bucket and not the pipes. Afterward, there is a rubber tubing connected to the bottom of the bucket, attached to the hole we cut out for drainage. There is also an emergency drainage when the bucket is overflowing after an extended period of time when it is not cleaned.

Advantages: The device collects a large amount of water, allowing the device to be left alone until hours later. There is a filter on the bottom; this allows the device to be cleared of debris when the filter is taken out, this allows the device to be cleaned more easily. The device will be underneath the AC, not blocking the walkway where the students would be walking.

Disadvantages: The design allows more debris to enter inside the device, allowing it to clog up and let water build up until it reaches its other drainage system and potentially leak, making the device useless. The device also takes a long time to assemble when transporting it. The PVC parts that will have to screw in take 3-5 minutes on average when assembling and disassembling the device.

Concept 3: Drain Pan



Description:

The Drain Pan receives the condensate from the AC units through a $\frac{1}{2}$ in diameter and 3 ft long rubber pipe. The drain pan stores at least 0.8 ft³ of water, and water needs to be removed every 5 hours.

Advantages:

It is easy to install; you just have to build the device first and easily put it under the AC and install the rubber tubing where the AC is dripping from. Debris will not be able to interfere with the device since the tubing on the device is installed into the AC and goes straight into the drain pan.

Disadvantages:

The Drain Pan design can only hold up to a certain amount of water. When it is collecting water, it will mean that the water will be building up in the rubber tubing and potentially overflow; this does not allow the device to be left alone for long periods of time, meaning it will have to be checked on frequently. It is also bigger than the design that we had originally aimed for, blocking some of the sidewalk. When transporting the device, it will become difficult to move it and dump the water in a certain location.

D3. PLAN OF ACTION

During the first week of starting the build of this device, we decided that the whole group would get the materials needed to build this device. We will each buy or let the teachers buy super glue, a transparent 1" pipe, and one of us will CAD the 3D parts that will be needed in this device, allowing for custom piping for the connection between the AC and piping to be connected. Ensuring that half of the material will be gained and making sure it is the right material that is needed before we build the device. In the second week, for the first two days, we decided that we would wait for the 3D printed parts to be printed, allowing us to make sure that all the parts needed were gained to ensure the making of this device. For the last two days of the second week, we decided to build the device, assembling the transparent tubing with the 3D printed parts with the super glue that is provided. In the first half of week three, we will install the device into the AC, enabling us to observe how the device works and how we can improve it if there are any problems with the device during the making of it.

D4. DESIGN BLUEPRINT

When making this device, you'll need to buy several things and 3D model some parts in order to make the device. Using Onshape, you will have to model four parts on Onshape, CAD the pieces out, and 3D print out those pieces. While printing those out while you're waiting, you can go to CityMill, find a transparent tubing that is 10ft long, and find super glue. Once you gather all the materials- the tubing, super glue, and the four parts that you've 3D printed out- you will need to assemble them. One side of the U-shape rubber tube will be installed onto the cross-shape plastic part, which will be connected to a 0.5 inches wide rubber tube that will later lead condensed water from the AC units. The other side of the U-shape will later be installed onto a 1.5 in long plastic pipe. Install the device onto an AC unit by connecting the 0.5 in wide rubber tube into the leaking hole on it, then stabilize the device by gluing part of it onto the AC unit, tape the parts together until dry, then untape it and install it onto the device.

Rationale:

Based on our decision matrix, we decided to choose concept design #1, Drain Trap. It provided the most benefits compared to the other designs that were also scored on that matrix. The design offered the most capacity without spilling the most compared to the other designs. It can also be built in under a minute, while needing more than 2 square feet. While design #1 only failed one of the less important attributes, other solutions such as design #2 take too long to construct and deconstruct, impose potential leaks, and allow debris to easily enter the device. On the other hand, design #3 also did not get a high score in the evaluation process because it might be inconvenient to carry due to its weight and structure, and just like design #2, it's an open container without blockage to prevent contaminants from entering. Therefore, with design #1 getting the highest evaluation score, it was chosen as the solution we decided to dive deep into.